



BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 3 Lab Packet.

Muscle · Head, Neck, Trunk, and Upper Limb Muscles · Heart · Vessels · Nerves · Blood

PART A

Lab structure list

Every structure you can be asked to identify on the Module 3 lab exam, chapter by chapter.

PART B

Muscle charts

Origin, insertion, action, and nerve for every muscle in this unit, including the face and neck.

PART C

Blood cell charts

What each cell looks like, its cell line, its job, when it runs high or low, and the terms for each.

PART D

Lab exam questions

The kinds of questions you could be asked at a station, by chapter.

MODULE 3 · LAB

How to use this packet In lab

Bring Part A to every lab. Check off a structure only when you can find it on the model, the cadaver, or the slide without looking at a label. Say the name out loud to your team as you point to it.

Studying the muscles

For every muscle, the lab exam can ask four things: where it starts (origin), where it ends (insertion), what it does (action), and which nerve supplies it. Cover the chart and quiz yourself one column at a time.



SCAN TO DRILL THE MUSCLES

Interactive Muscle Charts: I, O, A

Origins, insertions, actions, and innervation, drilled interactively. Use it on your phone between lab days.

medmasterscollaborative.com/muscle-charts-i-o-a-inn

PART A

Lab structure list.

These are the lab structures and concepts to learn for the lab portion of the Module 3 exam. Be prepared to identify structures on models, images, the cadaver, and slides.

CHAPTER 1

Muscle Tissue and Microanatomy

SLIDES AND THE SARCOMERE MODEL

Three tissues on the slide, then the sarcomere on the model. Fill in function and location for each tissue as you go; those columns are what the practical asks you to reason from.

☐ Skeletal muscle, longitudinal section

Function _____ Location _____

Special structures _____

STRUCTURES TO IDENTIFY

- | | | |
|---------------------------------------|--|-------------------------------------|
| ☐ Muscle fiber | ☐ Peripheral nuclei | ☐ Endomysium |
| ☐ Sarcolemma | ☐ Striations | |
| ☐ Sarcoplasm | ☐ Myofibrils | |

☐ Skeletal muscle, cross section

STRUCTURES TO IDENTIFY

- | | | |
|---|-------------------------------------|--|
| ☐ Muscle fiber, cut across | ☐ Endomysium | ☐ Epimysium |
| ☐ Fascicle | ☐ Perimysium | ☐ Peripheral nuclei |

☐ Cardiac muscle

Function _____ Location _____

Special structures _____

STRUCTURES TO IDENTIFY

- | | | |
|---|---|-------------------------------------|
| ☐ Cardiomyocyte | ☐ Central nucleus | ☐ Striations |
| ☐ Branching fibers | ☐ Intercalated discs | |

☐ Smooth muscle

Function _____ Location _____

Special structures _____

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Spindle-shaped cell | ☐ Absence of striations | ☐ Circular layer |
| ☐ Central nucleus | ☐ Longitudinal layer | |

☐ The muscle organ model

STRUCTURES TO IDENTIFY

- | | | |
|-------------------------------------|---------------------------------------|------------------------------------|
| ☐ Epimysium | ☐ Muscle fiber | ☐ Belly |
| ☐ Perimysium | ☐ Tendon | ☐ Origin |
| ☐ Endomysium | ☐ Aponeurosis | ☐ Insertion |
| ☐ Fascicle | ☐ Deep fascia | |

☐ The muscle fiber model

STRUCTURES TO IDENTIFY

- | | | |
|-------------------------------------|---|---|
| ☐ Sarcolemma | ☐ Myofibril | ☐ Terminal cisternae |
| ☐ Sarcoplasm | ☐ Transverse (T) tubule | ☐ Triad |
| ☐ Myonuclei | ☐ Sarcoplasmic reticulum | ☐ Mitochondria |

☐ The sarcomere model

Be prepared to state which filaments are present in each band, line, and zone.

STRUCTURES TO IDENTIFY

- | | | |
|---------------------------------|--|--|
| ☐ Z disc | ☐ M line | ☐ Titin |
| ☐ A band | ☐ Zone of overlap | ☐ Cross bridges |
| ☐ I band | ☐ Thick filament (myosin) | |
| ☐ H zone | ☐ Thin filament (actin) | |

Which regions of the sarcomere narrow when a muscle shortens?

Which region never changes length, and why?

☐ Fascicle arrangement, on models

Name one example muscle for each pattern.

STRUCTURES TO IDENTIFY

- | | | |
|-----------------------------------|-------------------------------------|---------------------------------------|
| ☐ Parallel | ☐ Convergent | ☐ Multipennate |
| ☐ Fusiform | ☐ Unipennate | |
| ☐ Circular | ☐ Bipennate | |

☐ Lever systems

Be prepared to classify a body movement by lever class and to name the fulcrum, effort, and load in it.

STRUCTURES TO IDENTIFY

- | | | |
|----------------------------------|--|---|
| ☐ Fulcrum | ☐ Load | ☐ Second class lever |
| ☐ Effort | ☐ First class lever | ☐ Third class lever |

CHAPTER 2

Muscles of the Head and Neck

MODELS AND CADAVER

Facial expression, chewing, and the neck. The extraocular muscles of the eye are not on this exam. For every muscle, be prepared to give the origin, the insertion, the action, and the nerve.

☐ Muscles of facial expression

All supplied by the facial nerve (CN VII).

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Occipitofrontalis, frontal belly | ☐ Nasalis | ☐ Risorius |
| ☐ Occipitofrontalis, occipital belly | ☐ Orbicularis oris | ☐ Buccinator |
| ☐ Galea aponeurotica (epicranial aponeurosis) | ☐ Zygomaticus major | ☐ Depressor anguli oris |
| ☐ Orbicularis oculi | ☐ Zygomaticus minor | ☐ Depressor labii inferioris |
| ☐ Procerus | ☐ Levator labii superioris | ☐ Mentalis |
| | ☐ Levator anguli oris | ☐ Platysma |

☐ Muscles of mastication

All supplied by the mandibular division of the trigeminal nerve (CN V3).

STRUCTURES TO IDENTIFY

- | | |
|-------------------------------------|--|
| ☐ Temporalis | ☐ Medial pterygoid |
| ☐ Masseter | ☐ Lateral pterygoid |

☐ Muscles of the neck

Use the hyoid bone and the thyroid cartilage as your landmarks.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Sternocleidomastoid | ☐ Mylohyoid | ☐ Omohyoid |
| ☐ Anterior scalene | ☐ Stylohyoid | ☐ Thyrohyoid |
| ☐ Middle scalene | ☐ Geniohyoid | ☐ Hyoid bone |
| ☐ Posterior scalene | ☐ Sternohyoid | ☐ Thyroid cartilage |
| ☐ Digastric | ☐ Sternothyroid | |

Which cranial nerve supplies every muscle of facial expression, and which supplies the muscles of mastication?

CHAPTER 3

Muscles of the Back and Thorax

TRUNK MUSCULATURE

Work these in layers, superficial to deep. For every muscle be prepared to give the origin, the insertion, and the action, not just the name.

☐ Superficial back

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Trapezius | ☐ Rhomboid major | ☐ Ligamentum nuchae |
| ☐ Latissimus dorsi | ☐ Rhomboid minor | |
| ☐ Levator scapulae | ☐ Thoracolumbar fascia | |

☐ Erector spinae

Identify them in order from lateral to medial. Be prepared to name the column from its position alone.

STRUCTURES TO IDENTIFY

- | | | |
|---------------------------------------|--------------------------------------|-----------------------------------|
| ☐ Iliocostalis | ☐ Longissimus | ☐ Spinalis |
|---------------------------------------|--------------------------------------|-----------------------------------|

☐ Deep back

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Splenius capitis | ☐ Multifidus | ☐ Interspinales |
| ☐ Splenius cervicis | ☐ Quadratus lumborum | ☐ Intertransversarii |
| ☐ Semispinalis | ☐ Rotatores | |

☐ Thoracic wall

Be prepared to state the fiber direction of the external and internal intercostals and what each does to the ribs.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|--|
| ☐ External intercostals | ☐ Transversus thoracis | ☐ Intercostal space |
| ☐ Internal intercostals | ☐ Serratus posterior superior | ☐ Neurovascular bundle in the costal groove |
| ☐ Innermost intercostals | ☐ Serratus posterior inferior | |

☐ The diaphragm

Name what passes through each of the three openings.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Central tendon | ☐ Costal part | ☐ Esophageal hiatus (T10) |
| ☐ Right crus | ☐ Sternal part | ☐ Aortic hiatus (T12) |
| ☐ Left crus | ☐ Caval opening (T8) | ☐ Phrenic nerve |

☐ Anterior thorax

STRUCTURES TO IDENTIFY

- | | | |
|---|--|--|
| ☐ Pectoralis major | ☐ Serratus anterior | ☐ External oblique, upper slips |
| ☐ Pectoralis minor | ☐ Subclavius | ☐ Clavipectoral fascia |

Which nerve supplies serratus anterior, and what deficit follows its injury?

CHAPTER 4

Muscles of the Upper Extremity

SHOULDER, ARM, FOREARM, HAND

Learn these by compartment. On a cadaver or a model the compartment tells you the action and the nerve before you have identified the individual muscle.

☐ Rotator cuff

Name the tubercle each one reaches and the movement each one produces.

STRUCTURES TO IDENTIFY

- | | |
|--|--|
| ☐ Supraspinatus | ☐ Teres minor |
| ☐ Infraspinatus | ☐ Subscapularis |

☐ Scapulohumeral, other

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Deltoid, anterior fibers | ☐ Deltoid, posterior fibers | ☐ Coracobrachialis |
| ☐ Deltoid, middle fibers | ☐ Teres major | |

☐ Arm, anterior compartment

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Biceps brachii, long head | ☐ Bicipital aponeurosis | ☐ Coracobrachialis |
| ☐ Biceps brachii, short head | ☐ Brachialis | |

☐ Arm, posterior compartment

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Triceps brachii, long head | ☐ Triceps brachii, medial head | ☐ Radial groove |
| ☐ Triceps brachii, lateral head | ☐ Anconeus | |

☐ Forearm, anterior compartment

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Brachioradialis | ☐ Flexor carpi ulnaris | ☐ Pronator quadratus |
| ☐ Pronator teres | ☐ Flexor digitorum superficialis | ☐ Medial epicondyle, common flexor origin |
| ☐ Flexor carpi radialis | ☐ Flexor digitorum profundus | |
| ☐ Palmaris longus | ☐ Flexor pollicis longus | |

☐ Forearm, posterior compartment

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Extensor carpi radialis longus | ☐ Extensor carpi ulnaris | ☐ Extensor pollicis longus |
| ☐ Extensor carpi radialis brevis | ☐ Supinator | ☐ Extensor indicis |
| ☐ Extensor digitorum | ☐ Abductor pollicis longus | ☐ Lateral epicondyle, common extensor origin |
| ☐ Extensor digiti minimi | ☐ Extensor pollicis brevis | |

☐ Hand, intrinsic muscles

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Thenar eminence | ☐ Adductor pollicis | ☐ Opponens digiti minimi |
| ☐ Abductor pollicis brevis | ☐ Hypothenar eminence | ☐ Lumbricals |
| ☐ Flexor pollicis brevis | ☐ Abductor digiti minimi | ☐ Palmar interossei |
| ☐ Opponens pollicis | ☐ Flexor digiti minimi brevis | ☐ Dorsal interossei |

☐ Landmarks and retinacula

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Cubital fossa | ☐ Extensor retinaculum | ☐ Extensor expansion |
| ☐ Carpal tunnel | ☐ Palmar aponeurosis | |
| ☐ Flexor retinaculum | ☐ Anatomical snuffbox | |

Name the two common origins that most superficial forearm muscles share.

CHAPTER 5

The Heart

MODEL, CADAVER, AND SLIDE

Work the heart from the outside in: sac, wall, chambers, valves, then the vessels on its surface. Be prepared to trace blood through it in order on any specimen you are given.

☐ Position and coverings

STRUCTURES TO IDENTIFY

- | | | |
|--------------------------------------|---|---|
| ☐ Mediastinum | ☐ Anterior (sternocostal) surface | ☐ Parietal serous pericardium |
| ☐ Apex | ☐ Inferior (diaphragmatic) surface | ☐ Visceral serous pericardium (epicardium) |
| ☐ Base | ☐ Fibrous pericardium | ☐ Pericardial cavity |

☐ The heart wall

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- | | | |
|-------------------------------------|-------------------------------------|--------------------------------------|
| ☐ Epicardium | ☐ Myocardium | ☐ Endocardium |
|-------------------------------------|-------------------------------------|--------------------------------------|

☐ External surface features

STRUCTURES TO IDENTIFY

- | | | |
|---|--|--|
| ☐ Coronary sulcus | ☐ Posterior interventricular sulcus | ☐ Left auricle |
| ☐ Anterior interventricular sulcus | ☐ Right auricle | ☐ Ligamentum arteriosum |

☐ Chambers and internal features

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Right atrium | ☐ Fossa ovalis | ☐ Chordae tendineae |
| ☐ Right ventricle | ☐ Pectinate muscles | ☐ Moderator band |
| ☐ Left atrium | ☐ Crista terminalis | ☐ Conus arteriosus |
| ☐ Left ventricle | ☐ Sinus venarum | ☐ Aortic vestibule |
| ☐ Interatrial septum | ☐ Trabeculae carneae | |
| ☐ Interventricular septum | ☐ Papillary muscles | |

☐ Valves

For each valve, name what it separates and what it prevents.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Tricuspid valve | ☐ Pulmonary semilunar valve | ☐ Fibrous skeleton |
| ☐ Mitral (bicuspid) valve | ☐ Aortic semilunar valve | |

☐ Great vessels

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Superior vena cava | ☐ Right pulmonary artery | ☐ Ascending aorta |
| ☐ Inferior vena cava | ☐ Left pulmonary artery | ☐ Aortic arch |
| ☐ Pulmonary trunk | ☐ Pulmonary veins | ☐ Descending thoracic aorta |

☐ Coronary circulation

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Right coronary artery | ☐ Anterior interventricular artery (LAD) | ☐ Great cardiac vein |
| ☐ Right marginal artery | ☐ Circumflex artery | ☐ Middle cardiac vein |
| ☐ Posterior interventricular artery (PDA) | ☐ Left marginal artery | ☐ Small cardiac vein |
| ☐ Left coronary artery | ☐ Coronary sinus | |

☐ **Conduction system, on the model**

STRUCTURES TO IDENTIFY

- | | | |
|----------------------------------|--|---|
| ☐ SA node | ☐ AV bundle (bundle of His) | ☐ Left bundle branch |
| ☐ AV node | ☐ Right bundle branch | ☐ Purkinje fibers |

☐ **Cardiac muscle slide**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- | | | |
|--|---|-------------------------------------|
| ☐ Cardiomyocyte | ☐ Branching fibers | ☐ Striations |
| ☐ Central nucleus | ☐ Intercalated discs | |

Trace a drop of blood from the superior vena cava to the aorta, naming every chamber and valve.

CHAPTER 6

Blood Vessels, Histology

SLIDES AND WALL MODELS

Almost every question here is answered from the wall. Identify the tunics first, then use their proportions to name the vessel.

☐ **Artery, cross section**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---------------------------------------|
| ☐ Tunica intima | ☐ Tunica media | ☐ Vasa vasorum |
| ☐ Endothelium | ☐ Smooth muscle | ☐ Lumen |
| ☐ Basement membrane | ☐ External elastic lamina | |
| ☐ Internal elastic lamina | ☐ Tunica externa | |

☐ **Vein, cross section**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--------------------------------|
| ☐ Tunica intima | ☐ Tunica externa | ☐ Lumen |
| ☐ Tunica media | ☐ Venous valve | |

Name the two features you would use first to tell an artery from a vein on a slide.

☐ Elastic artery

Name two elastic arteries you would expect to see labelled.

STRUCTURES TO IDENTIFY

- | | |
|---|--|
| ☐ Elastic lamellae | ☐ Internal elastic lamina |
| ☐ Thick tunica media | ☐ External elastic lamina |

☐ Muscular artery

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Prominent internal elastic lamina | ☐ Smooth muscle of the tunica media | ☐ Thin external elastic lamina |
|--|--|---|

☐ Arteriole and capillary bed

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Arteriole | ☐ Capillary | ☐ Postcapillary venule |
| ☐ Metarteriole | ☐ Capillary bed | ☐ Muscular venule |
| ☐ Precapillary sphincter | ☐ Thoroughfare channel | |

☐ Capillary types

For each type, name one organ where it is found and state what can cross its wall.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Continuous capillary | ☐ Sinusoid | ☐ Fenestration |
| ☐ Fenestrated capillary | ☐ Intercellular cleft | ☐ Basement membrane |

☐ Fetal circulation, on the model

Name the adult remnant of each fetal structure.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Placenta | ☐ Ductus venosus | ☐ Ductus arteriosus |
| ☐ Umbilical vein | ☐ Foramen ovale | ☐ Umbilical arteries |

CHAPTER 7

Arteries and Veins of the Thorax and Upper Limb

NAMED VESSELS, CADAVER AND MODEL

These are routes. Trace each one continuously with your finger before you try to name segments, and learn the landmark where each name changes.

☐ Aortic arch and its branches

Be prepared to explain why the right and left sides are not symmetrical.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Ascending aorta | ☐ Right common carotid artery | ☐ Left subclavian artery |
| ☐ Aortic arch | ☐ Right subclavian artery | ☐ Descending thoracic aorta |
| ☐ Brachiocephalic trunk | ☐ Left common carotid artery | |

☐ Other thoracic arteries

STRUCTURES TO IDENTIFY

- | | | |
|---|--|--|
| ☐ Internal thoracic artery | ☐ Bronchial arteries | ☐ Thyrocervical trunk |
| ☐ Anterior intercostal arteries | ☐ Esophageal arteries | ☐ Costocervical trunk |
| ☐ Posterior intercostal arteries | ☐ Vertebral artery | |

☐ Arteries of the upper limb

Name the landmark at which the subclavian becomes the axillary, and the axillary becomes the brachial.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Subclavian artery | ☐ Anterior circumflex humeral artery | ☐ Ulnar artery |
| ☐ Axillary artery | ☐ Posterior circumflex humeral artery | ☐ Common interosseous artery |
| ☐ Thoracoacromial artery | ☐ Brachial artery | ☐ Superficial palmar arch |
| ☐ Lateral thoracic artery | ☐ Deep brachial artery | ☐ Deep palmar arch |
| ☐ Subscapular artery | ☐ Radial artery | ☐ Digital arteries |

☐ Superficial veins of the upper limb

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Cephalic vein | ☐ Median cubital vein | ☐ Dorsal venous network of the hand |
| ☐ Basilic vein | ☐ Median antebrachial vein | |

☐ Deep veins of the thorax and upper limb

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Radial veins | ☐ Subclavian vein | ☐ Azygos vein |
| ☐ Ulnar veins | ☐ Internal jugular vein | ☐ Hemiazygos vein |
| ☐ Brachial veins | ☐ Brachiocephalic veins | ☐ Internal thoracic vein |
| ☐ Axillary vein | ☐ Superior vena cava | |

Why is there one brachiocephalic artery but two brachiocephalic veins?

CHAPTER 8

Nerves of the Upper Limb

THE BRACHIAL PLEXUS

Build the plexus by level before you learn individual nerves. The counts, five roots, three trunks, six divisions, three cords, five branches, are how you check your own work.

☐ Levels of the plexus

The cords are named for their position around the axillary artery. Identify the artery first.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Roots, C5 to T1 | ☐ Lower trunk | ☐ Lateral cord |
| ☐ Upper trunk | ☐ Anterior divisions | ☐ Posterior cord |
| ☐ Middle trunk | ☐ Posterior divisions | ☐ Medial cord |

☐ Terminal branches

For each nerve, name the compartment it supplies and the deficit its injury produces.

STRUCTURES TO IDENTIFY

- | | | |
|---|---------------------------------------|---|
| ☐ Musculocutaneous nerve | ☐ Ulnar nerve | ☐ Axillary nerve |
| ☐ Median nerve | ☐ Radial nerve | |

☐ **Branches above the terminal nerves**

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Dorsal scapular nerve | ☐ Lateral pectoral nerve | ☐ Lower subscapular nerve |
| ☐ Long thoracic nerve | ☐ Medial pectoral nerve | ☐ Medial cutaneous nerve of the arm |
| ☐ Suprascapular nerve | ☐ Thoracodorsal nerve | ☐ Medial cutaneous nerve of the forearm |
| ☐ Nerve to subclavius | ☐ Upper subscapular nerve | |

☐ **Nerves at their vulnerable points**

Name the injury associated with each site and the sign it produces.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Axillary nerve at the surgical neck of the humerus | ☐ Ulnar nerve behind the medial epicondyle | ☐ Long thoracic nerve on the chest wall |
| ☐ Radial nerve in the radial groove | ☐ Median nerve in the carpal tunnel | |

☐ **Thoracic nerves**

STRUCTURES TO IDENTIFY

- | | |
|---|--|
| ☐ Intercostal nerves | ☐ Vagus nerve in the thorax |
| ☐ Phrenic nerve | ☐ Sympathetic trunk |

CHAPTER 9

Blood

THE SMEAR

One slide, and every question comes off the nucleus. Identify each cell by the shape of its nucleus first and the granules second.

☐ **Whole blood, in a spun tube**

STRUCTURES TO IDENTIFY

- | | |
|---------------------------------------|--|
| ☐ Plasma layer | ☐ Erythrocyte layer |
| ☐ Buffy coat | ☐ Hematocrit |

☐ **Erythrocytes**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- | | |
|--|---|
| ☐ Biconcave disc | ☐ Absence of a nucleus |
| ☐ Pale central area | ☐ Rouleaux formation |

☐ **Granulocytes**

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Neutrophil | ☐ Eosinophil | ☐ Basophil |
| ☐ Multi-lobed nucleus | ☐ Bilobed nucleus | ☐ Nucleus obscured by dark granules |
| ☐ Fine pale granules | ☐ Coarse red-orange granules | |

☐ **Agranulocytes**

STRUCTURES TO IDENTIFY

- | | | |
|---|--|--|
| ☐ Lymphocyte | ☐ Monocyte | ☐ Abundant pale cytoplasm |
| ☐ Large round nucleus with thin rim of cytoplasm | ☐ Kidney-shaped nucleus | |

☐ **Platelets and formation**

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---------------------------------------|
| ☐ Platelet (thrombocyte) | ☐ Red bone marrow | ☐ Reticulocyte |
| ☐ Megakaryocyte | ☐ Hemocytoblast | |

Order the five leukocytes from most to least numerous.

PART B

Muscle charts.

Origin, insertion, action, and nerve for every muscle in Module 3, grouped the way you will find them in lab. The same information is in the interactive muscle charts.



SCAN TO DRILL THESE CHARTS

Interactive Muscle Charts: I, O, A

Quiz yourself on the origin, insertion, action, and nerve for each muscle on your phone.

medmasterscollaborative.com/muscle-charts-i-o-a-inn

Muscles of facial expression

Every muscle of facial expression is supplied by the facial nerve (CN VII). They pull on skin, not on bone, which is why they can make expressions.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Occipitofrontalis, frontal belly	Galea aponeurotica (epicranial aponeurosis)	Skin of the eyebrows and forehead	Raises the eyebrows and wrinkles the forehead	Facial nerve (CN VII)
Occipitofrontalis, occipital belly	Superior nuchal line of the occipital bone and the mastoid region	Galea aponeurotica	Pulls the scalp backward	Facial nerve (CN VII)
Orbicularis oculi	Medial margin of the orbit (frontal bone, maxilla) and the medial palpebral ligament	Skin around the eyelids	Closes the eye, as in blinking and squinting	Facial nerve (CN VII)
Procerus	Fascia over the nasal bone	Skin between the eyebrows	Pulls the medial eyebrows down and wrinkles the bridge of the nose	Facial nerve (CN VII)
Nasalis	Maxilla beside the nose	Nasal cartilages and the bridge of the nose	Compresses the nasal opening and flares the nostrils	Facial nerve (CN VII)
Orbicularis oris	Maxilla, mandible, and the muscles that blend in at the corners of the mouth	Skin and mucosa of the lips	Closes and purses the lips	Facial nerve (CN VII)
Zygomaticus major	Zygomatic bone, lateral surface	Angle (corner) of the mouth	Draws the corner of the mouth up and out, as in smiling	Facial nerve (CN VII)
Zygomaticus minor	Zygomatic bone	Upper lip	Elevates the upper lip	Facial nerve (CN VII)
Levator labii superioris	Maxilla, along the infraorbital margin	Upper lip	Elevates the upper lip	Facial nerve (CN VII)
Levator anguli oris	Maxilla, below the infraorbital foramen	Angle of the mouth	Elevates the corner of the mouth	Facial nerve (CN VII)
Risorius	Fascia over the masseter	Angle of the mouth	Draws the corner of the mouth sideways	Facial nerve (CN VII)
Buccinator	Alveolar processes of the maxilla and mandible in the molar region	Orbicularis oris at the angle of the mouth	Presses the cheek against the teeth, keeping food between them while chewing	Facial nerve (CN VII)

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Depressor anguli oris	Body of the mandible (oblique line)	Angle of the mouth	Pulls the corner of the mouth down, as in frowning	Facial nerve (CN VII)
Depressor labii inferioris	Body of the mandible, below the lower lip	Lower lip	Pulls the lower lip down	Facial nerve (CN VII)
Mentalis	Mandible, below the incisors	Skin of the chin	Raises and wrinkles the chin and pushes out the lower lip	Facial nerve (CN VII)
Platysma	Fascia over the upper pectoralis major and deltoid	Lower border of the mandible and the skin at the corner of the mouth	Tenses the skin of the neck and pulls the lower lip and corner of the mouth down	Facial nerve (CN VII), cervical branch

Muscles of mastication

All four chewing muscles are supplied by the mandibular division of the trigeminal nerve (CN V3).

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Temporalis	Temporal fossa of the skull	Coronoid process of the mandible	Elevates (closes) and retracts the mandible	Trigeminal nerve, mandibular division (CN V3)
Masseter	Zygomatic arch	Lateral ramus and angle of the mandible	Elevates the mandible; the prime mover for closing the jaw	Trigeminal nerve, mandibular division (CN V3)
Medial pterygoid	Medial surface of the lateral pterygoid plate of the sphenoid	Medial surface of the ramus and angle of the mandible	Elevates and protrudes the mandible; side to side grinding	Trigeminal nerve, mandibular division (CN V3)
Lateral pterygoid	Greater wing of the sphenoid and the lateral surface of the lateral pterygoid plate	Neck of the mandible and the disc of the temporomandibular joint	Protrudes the mandible and opens the jaw; side to side grinding	Trigeminal nerve, mandibular division (CN V3)

Muscles of the neck

The hyoid bone sorts these. Suprahyoid muscles sit above it and elevate it. Infrahyoid muscles sit below it and depress it.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Sternocleidomastoid	Manubrium of the sternum and the medial clavicle	Mastoid process of the temporal bone	Both sides: flex the neck. One side: tilts the head to the same side and turns the face to the opposite side	Accessory nerve (CN XI)
Scalenes (anterior, middle, posterior)	Transverse processes of the cervical vertebrae	Rib 1 (anterior and middle) and rib 2 (posterior)	Elevate ribs 1 and 2 and flex the neck to the same side	Ventral rami of the cervical spinal nerves
Digastric	Anterior belly: mandible near the midline. Posterior belly: mastoid notch of the temporal bone	Intermediate tendon held to the hyoid by a fibrous loop	Elevates the hyoid and depresses the mandible	Anterior belly: CN V3 (nerve to mylohyoid). Posterior belly: facial nerve (CN VII)
Mylohyoid	Mylohyoid line on the inside of the mandible	Body of the hyoid	Elevates the hyoid and forms the floor of the mouth	Trigeminal nerve, CN V3 (nerve to mylohyoid)
Stylohyoid	Styloid process of the temporal bone	Body of the hyoid	Elevates and retracts the hyoid	Facial nerve (CN VII)
Geniohyoid	Inner surface of the mandible at the chin	Body of the hyoid	Pulls the hyoid up and forward	C1 fibers traveling with the hypoglossal nerve (CN XII)
Sternohyoid	Manubrium and the medial clavicle	Body of the hyoid	Depresses the hyoid	Ansa cervicalis (C1 to C3)
Sternothyroid	Posterior surface of the manubrium	Thyroid cartilage	Depresses the larynx	Ansa cervicalis (C1 to C3)

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Omohyoid	Superior border of the scapula	Body of the hyoid (two bellies joined by an intermediate tendon)	Depresses the hyoid	Ansa cervicalis (C1 to C3)
Thyrohyoid	Thyroid cartilage	Body and greater horn of the hyoid	Depresses the hyoid and elevates the larynx	C1 fibers traveling with the hypoglossal nerve (CN XII)

Superficial muscles of the back

These move the scapula and the arm. All but the trapezius get their main nerve supply from branches of the brachial plexus (levator scapulae also receives C3 and C4).

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Trapezius	Occipital bone, ligamentum nuchae, spinous processes C7 to T12	Lateral clavicle, acromion, spine of the scapula	Upper fibers elevate the scapula, middle fibers retract it, lower fibers depress it; upper and lower fibers together rotate the glenoid upward	Accessory nerve (CN XI)
Latissimus dorsi	Spinous processes T7 to L5, thoracolumbar fascia, iliac crest, lower ribs	Intertubercular sulcus of the humerus	Extends, adducts, and medially rotates the arm	Thoracodorsal nerve
Levator scapulae	Transverse processes C1 to C4	Superior medial border of the scapula	Elevates the scapula and rotates the glenoid downward	Dorsal scapular nerve and cervical nerves (C3, C4)
Rhomboid major	Spinous processes T2 to T5	Medial border of the scapula, below the spine	Retracts the scapula and rotates the glenoid downward	Dorsal scapular nerve
Rhomboid minor	Spinous processes C7 to T1	Medial border of the scapula, at the spine	Retracts the scapula and rotates the glenoid downward	Dorsal scapular nerve

Deep muscles of the back

The deep back muscles are all supplied by the dorsal rami of the spinal nerves at their levels, except the quadratus lumborum, and some intertransversarii also receive ventral rami.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Iliocostalis (erector spinae, lateral column)	Iliac crest, sacrum, and the angles of the lower ribs	Angles of the ribs and the transverse processes of the lower cervical vertebrae	Both sides: extend the vertebral column. One side: bends it to the same side	Dorsal rami of the spinal nerves
Longissimus (erector spinae, intermediate column)	Transverse processes of the lumbar, thoracic, and lower cervical vertebrae	Transverse processes above, the ribs, and the mastoid process	Extends the vertebral column and the head; bends to the same side	Dorsal rami of the spinal nerves
Spinalis (erector spinae, medial column)	Spinous processes of the upper lumbar and lower thoracic vertebrae	Spinous processes of the upper thoracic and cervical vertebrae	Extends the vertebral column	Dorsal rami of the spinal nerves
Splenius capitis	Ligamentum nuchae, spinous processes C7 to T3	Mastoid process and occipital bone	Together extend the head; singly rotate the head to the same side	Dorsal rami of the cervical spinal nerves
Splenius cervicis	Spinous processes T3 to T6	Transverse processes C1 to C3	Extends and rotates the neck	Dorsal rami of the cervical spinal nerves
Semispinalis	Transverse processes C7 to T12	Occipital bone and spinous processes above	Extends the head and vertebral column	Dorsal rami of the spinal nerves
Multifidus	Sacrum, ilium, transverse processes	Spinous processes two to four segments above	Extends and rotates the vertebral column, and stabilises it segment by segment	Dorsal rami of the spinal nerves

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Rotatores	Transverse process of a vertebra	Lamina and spinous process one or two vertebrae above	Rotate and extend the vertebral column and stabilise it	Dorsal rami of the spinal nerves
Interspinales	Spinous process of a vertebra	Spinous process of the vertebra above	Extend the vertebral column	Dorsal rami of the spinal nerves
Intertransversarii	Transverse process of a vertebra	Transverse process of the vertebra above	Bend the vertebral column to the same side	Dorsal and ventral rami of the spinal nerves
Quadratus lumborum	Iliac crest and iliolumbar ligament	Rib 12 and transverse processes L1 to L4	Laterally flexes the trunk; both together extend the lumbar spine	Ventral rami of T12 to L4

Muscles of the thoracic wall and the diaphragm

The intercostal nerves run in the costal groove, between the internal and innermost intercostals, with the intercostal artery and vein.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
External intercostals	Inferior border of the rib above	Superior border of the rib below	Elevate the ribs. Fibers run downward and forward, hands in pockets	Intercostal nerves
Internal intercostals	Superior border of the rib below	Inferior border of the rib above	Depress the ribs. Fibers run at right angles to the external layer	Intercostal nerves
Innermost intercostals	Internal surface of the rib below	Internal surface of the rib above	Act with the internal intercostals; separated from them by the neurovascular bundle	Intercostal nerves
Transversus thoracis	Posterior surface of the lower sternum	Costal cartilages 2 to 6	Depresses the ribs	Intercostal nerves
Serratus posterior superior	Spinous processes C7 to T3	Ribs 2 to 5	Elevates the ribs	Intercostal nerves (T2 to T5)
Serratus posterior inferior	Spinous processes T11 to L2	Ribs 9 to 12	Depresses the ribs	Lower intercostal nerves and the subcostal nerve (T9 to T12)
Diaphragm	Xiphoid process, inner surface of ribs 7 to 12, and the lumbar vertebrae by two crura	Central tendon	Contracts and flattens, which increases the vertical dimension of the thoracic cavity	Phrenic nerve (C3, C4, C5)

Anterior thoracic muscles

These four act on the pectoral girdle or the arm from the front of the chest.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Pectoralis major	Medial clavicle, sternum, costal cartilages 1 to 6	Lateral lip of the intertubercular sulcus of the humerus	Flexes, adducts, and medially rotates the arm	Medial and lateral pectoral nerves
Pectoralis minor	Ribs 3 to 5	Coracoid process of the scapula	Depresses and protracts the scapula, and elevates the ribs when the scapula is fixed	Medial pectoral nerve
Serratus anterior	Ribs 1 to 8 or 9, on the lateral chest wall	Anterior surface of the medial border of the scapula	Protracts the scapula and rotates the glenoid upward; holds the scapula against the chest wall	Long thoracic nerve
Subclavius	Rib 1 at the costal cartilage	Inferior surface of the clavicle	Depresses and steadies the clavicle	Nerve to subclavius

Rotator cuff and other muscles crossing the shoulder

Supraspinatus, infraspinatus, and teres minor all reach the greater tubercle. Subscapularis is the only cuff muscle on the lesser tubercle.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Supraspinatus	Supraspinous fossa of the scapula	Greater tubercle of the humerus	Initiates abduction of the arm, the first fifteen degrees	Suprascapular nerve
Infraspinatus	Infraspinous fossa of the scapula	Greater tubercle of the humerus	Laterally rotates the arm	Suprascapular nerve
Teres minor	Lateral border of the scapula	Greater tubercle of the humerus	Laterally rotates the arm	Axillary nerve
Subscapularis	Subscapular fossa, the anterior surface of the scapula	Lesser tubercle of the humerus	Medially rotates the arm	Upper and lower subscapular nerves
Deltoid	Lateral clavicle, acromion, spine of the scapula	Deltoid tuberosity of the humerus	Anterior fibers flex and medially rotate, middle fibers abduct, posterior fibers extend and laterally rotate	Axillary nerve
Teres major	Inferior angle of the scapula	Medial lip of the intertubercular sulcus	Extends, adducts, and medially rotates the arm	Lower subscapular nerve
Coracobrachialis	Coracoid process of the scapula	Medial surface of the humeral shaft	Flexes and adducts the arm	Musculocutaneous nerve

Muscles of the arm

Anterior compartment: musculocutaneous nerve. Posterior compartment: radial nerve.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Biceps brachii	Long head from the supraglenoid tubercle, short head from the coracoid process	Radial tuberosity and the bicipital aponeurosis	Flexes the elbow and supinates the forearm; also flexes the arm	Musculocutaneous nerve
Brachialis	Distal anterior humerus	Coronoid process and tuberosity of the ulna	Flexes the elbow, the prime mover regardless of forearm position	Musculocutaneous nerve
Triceps brachii	Long head from the infraglenoid tubercle, lateral and medial heads from the humerus	Olecranon of the ulna	Extends the elbow; the long head also extends the arm	Radial nerve
Anconeus	Lateral epicondyle of the humerus	Lateral olecranon and posterior ulna	Assists extension of the elbow	Radial nerve

Anterior forearm (flexor compartment)

Most start at the medial epicondyle (common flexor origin). Median nerve, except flexor carpi ulnaris and the medial half of flexor digitorum profundus, which are ulnar.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Pronator teres	Medial epicondyle of the humerus and coronoid process of the ulna	Lateral mid-shaft of the radius	Pronates the forearm and assists elbow flexion	Median nerve
Flexor carpi radialis	Medial epicondyle	Bases of metacarpals 2 and 3	Flexes and abducts the wrist	Median nerve

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Palmaris longus	Medial epicondyle	Palmar aponeurosis	Tenses the palmar aponeurosis; absent in many people	Median nerve
Flexor carpi ulnaris	Medial epicondyle and olecranon	Pisiform, hamate, metacarpal 5	Flexes and adducts the wrist	Ulnar nerve
Flexor digitorum superficialis	Medial epicondyle, coronoid process, radius	Middle phalanges of digits 2 to 5	Flexes the proximal interphalangeal joints	Median nerve
Flexor digitorum profundus	Anterior and medial ulna and interosseous membrane	Distal phalanges of digits 2 to 5	Flexes the distal interphalangeal joints	Lateral half: median nerve. Medial half: ulnar nerve
Flexor pollicis longus	Anterior radius and interosseous membrane	Distal phalanx of the thumb	Flexes the thumb	Median nerve (anterior interosseous branch)
Pronator quadratus	Distal anterior ulna	Distal anterior radius	Pronates the forearm, the prime mover of pronation	Median nerve (anterior interosseous branch)

Posterior forearm (extensor compartment)

Most start at the lateral epicondyle (common extensor origin). Every muscle here is supplied by the radial nerve or its deep branch.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Brachioradialis	Lateral supracondylar ridge of the humerus	Styloid process of the radius	Flexes the elbow, strongest with the forearm in mid-position	Radial nerve
Extensor carpi radialis longus	Lateral supracondylar ridge	Base of metacarpal 2	Extends and abducts the wrist	Radial nerve
Extensor carpi radialis brevis	Lateral epicondyle	Base of metacarpal 3	Extends and abducts the wrist	Radial nerve, deep branch
Extensor digitorum	Lateral epicondyle	Extensor expansions of digits 2 to 5	Extends the fingers and the wrist	Radial nerve, deep branch (posterior interosseous)
Extensor digiti minimi	Lateral epicondyle	Extensor expansion of digit 5	Extends the little finger	Radial nerve, deep branch (posterior interosseous)
Extensor carpi ulnaris	Lateral epicondyle and posterior ulna	Base of metacarpal 5	Extends and adducts the wrist	Radial nerve, deep branch (posterior interosseous)
Supinator	Lateral epicondyle and proximal ulna	Proximal lateral radius	Supinates the forearm	Radial nerve, deep branch
Abductor pollicis longus	Posterior radius, ulna, interosseous membrane	Base of metacarpal 1	Abducts and extends the thumb	Radial nerve, deep branch (posterior interosseous)
Extensor pollicis brevis	Posterior radius and interosseous membrane	Proximal phalanx of the thumb	Extends the thumb at the metacarpophalangeal joint	Radial nerve, deep branch (posterior interosseous)
Extensor pollicis longus	Posterior ulna and interosseous membrane	Distal phalanx of the thumb	Extends the thumb at the interphalangeal joint	Radial nerve, deep branch (posterior interosseous)
Extensor indicis	Posterior ulna and interosseous membrane	Extensor expansion of digit 2	Extends the index finger	Radial nerve, deep branch (posterior interosseous)

Intrinsic muscles of the hand

The ulnar nerve supplies most of the hand. The median nerve supplies the thenar muscles (except adductor pollicis) and lumbricals 1 and 2.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Abductor pollicis brevis	Flexor retinaculum, scaphoid, and trapezium	Base of the proximal phalanx of the thumb	Abducts the thumb	Median nerve (recurrent branch)
Flexor pollicis brevis	Flexor retinaculum and trapezium	Base of the proximal phalanx of the thumb	Flexes the thumb at the metacarpophalangeal joint	Median nerve (recurrent branch); deep head, ulnar nerve
Opponens pollicis	Flexor retinaculum and trapezium	Lateral side of metacarpal 1	Opposes the thumb, bringing it across the palm to the fingers	Median nerve (recurrent branch)
Adductor pollicis	Capitate and metacarpals 2 and 3	Base of the proximal phalanx of the thumb, medial side	Adducts the thumb	Ulnar nerve, deep branch
Abductor digiti minimi	Pisiform	Base of the proximal phalanx of digit 5	Abducts the little finger	Ulnar nerve, deep branch
Flexor digiti minimi brevis	Hook of the hamate and flexor retinaculum	Base of the proximal phalanx of digit 5	Flexes the little finger at the metacarpophalangeal joint	Ulnar nerve, deep branch
Opponens digiti minimi	Hook of the hamate and flexor retinaculum	Medial side of metacarpal 5	Brings metacarpal 5 toward the thumb	Ulnar nerve, deep branch
Lumbricals	Tendons of flexor digitorum profundus	Lateral side of the extensor expansions of digits 2 to 5	Flex the metacarpophalangeal joints and extend the interphalangeal joints	Lumbricals 1 and 2: median nerve. Lumbricals 3 and 4: ulnar nerve
Palmar interossei	Metacarpals 2, 4, and 5	Proximal phalanges and extensor expansions of the same digits	Adduct the fingers toward the middle finger (PAD)	Ulnar nerve, deep branch
Dorsal interossei	Adjacent sides of metacarpals 1 to 5	Proximal phalanges and extensor expansions of digits 2 to 4	Abduct the fingers away from the middle finger (DAB)	Ulnar nerve, deep branch

PART C

Blood cell charts.

Identify every cell by the shape of its nucleus first and its granules second. The counts are typical adult ranges; labs vary slightly.

The formed elements, part 1: what it is

CELL	WHAT IT LOOKS LIKE	CELL LINE	GRANULOCYTE OR AGRANULOCYTE	HOW MANY	MAIN JOB
Erythrocyte (red blood cell)	Biconcave disc about 7.5 micrometers across with a pale center. No nucleus and no organelles; packed with hemoglobin. Can stack like coins (rouleaux).	Myeloid	Neither (not a leukocyte)	About 4 to 6 million per microliter; the most numerous formed element	Carries oxygen to the tissues and some carbon dioxide back, bound to hemoglobin
Neutrophil	Nucleus with three to five lobes joined by thin strands. Fine, pale lilac granules. About twice the size of a red cell.	Myeloid	Granulocyte	50 to 70 percent of white cells; the most numerous leukocyte	First responder. Leaves the blood and engulfs bacteria (phagocytosis)
Eosinophil	Nucleus with two lobes. Large red-orange granules that fill the cytoplasm.	Myeloid	Granulocyte	2 to 4 percent of white cells	Attacks parasitic worms and helps control allergic reactions
Basophil	Lobed nucleus that is usually hidden under large, dark blue-purple granules.	Myeloid	Granulocyte	Less than 1 percent of white cells; the least numerous leukocyte	Releases histamine, which widens blood vessels during inflammation and allergy, and heparin, which slows clotting
Lymphocyte	Large, round, dark-staining nucleus that fills most of the cell, with a thin rim of pale blue cytoplasm. Usually a little larger than a red cell.	Lymphoid	Agranulocyte	20 to 40 percent of white cells; the second most numerous leukocyte	Specific immunity. T cells attack infected cells directly; B cells become plasma cells that make antibodies
Monocyte	The largest white cell. Kidney- or horseshoe-shaped nucleus with abundant pale gray-blue cytoplasm.	Myeloid	Agranulocyte	2 to 8 percent of white cells	Leaves the blood and becomes a macrophage that engulfs pathogens and debris
Platelet (thrombocyte)	Small purple cell fragment, 2 to 4 micrometers, with no nucleus. Often seen in small clumps between the red cells.	Myeloid (breaks off a megakaryocyte)	Neither (a cell fragment)	150,000 to 400,000 per microliter	Forms the platelet plug and helps start clotting when a vessel is damaged

The formed elements, part 2: too many and too few

The term for a high count usually ends in -philia or -cytosis. The term for a low count ends in -penia.

CELL	ELEVATED WHEN	TERM	DECREASED WHEN	TERM
Erythrocyte (red blood cell)	Living at high altitude, chronic lung disease, dehydration, or a marrow disorder (polycythemia vera)	Polycythemia	Blood loss, iron or vitamin B12 deficiency, kidney disease, destruction of red cells, or marrow failure	Anemia
Neutrophil	Acute bacterial infection, inflammation, tissue injury, or stress	Neutrophilia	Chemotherapy, some viral infections, some drugs, or marrow suppression	Neutropenia
Eosinophil	Parasitic worm infection, allergies, and asthma	Eosinophilia	Corticosteroid drugs or high cortisol from stress	Eosinopenia

CELL	ELEVATED WHEN	TERM	DECREASED WHEN	TERM
Basophil	Allergic reactions and some leukemias, such as chronic myeloid leukemia	Basophilia	Corticosteroid drugs, stress, or an overactive thyroid (hard to measure because the normal count is so low)	Basopenia
Lymphocyte	Viral infections such as mononucleosis, and chronic lymphocytic leukemia	Lymphocytosis	HIV infection, corticosteroid drugs, chemotherapy, or radiation	Lymphopenia
Monocyte	Chronic infections such as tuberculosis, and chronic inflammation	Monocytosis	Corticosteroid drugs or marrow suppression	Monocytopenia
Platelet (thrombocyte)	Inflammation, iron deficiency, after the spleen is removed, or a marrow disorder	Thrombocytosis	Leukemia, chemotherapy, some drugs, viral infection, or an enlarged spleen that traps platelets	Thrombocytopenia

Where the cells come from

CELL	WHAT IT LOOKS LIKE	WHAT TO KNOW
Hemocytoblast	Hematopoietic stem cell in red bone marrow	Gives rise to both the myeloid and lymphoid lines, so every formed element starts here
Megakaryocyte	Very large marrow cell with a big, many-lobed nucleus	Breaks into thousands of platelets; stays in the red bone marrow
Reticulocyte	Young red cell that has just lost its nucleus; a network of leftover ribosomes shows with a special stain	About 1 percent of red cells. The count rises after blood loss as the marrow speeds up (reticulocytosis) and falls when the marrow fails

Memory line for the white cells from most to least numerous: neutrophil, lymphocyte, monocyte, eosinophil, basophil ("Never Let Monkeys Eat Bananas").

Blood terms

Hematocrit

The percent of blood volume made up of red cells after the blood is spun. Roughly 42 to 52 percent in males and 37 to 47 percent in females.

Plasma

The liquid part of blood, about 55 percent of the volume. Mostly water, plus proteins (albumin, globulins, fibrinogen), nutrients, gases, and wastes.

Hematopoiesis

Formation of all blood cells, in the red bone marrow. Erythropoiesis makes red cells, leukopoiesis makes white cells, and thrombopoiesis makes platelets.

Lymphoid line

The stem cell line that makes lymphocytes.

Agranulocyte

A white cell without visible granules: lymphocyte and monocyte.

Leukopenia

A white cell count below normal (below about 4,500 per microliter).

Buffy coat

The thin whitish layer between the plasma and the red cells in a spun tube. It holds the white cells and platelets, less than 1 percent of the volume.

Formed elements

The cells and cell fragments of blood: erythrocytes, leukocytes, and platelets.

Myeloid line

The stem cell line that makes red cells, platelets, neutrophils, eosinophils, basophils, and monocytes.

Granulocyte

A white cell with visible granules in its cytoplasm: neutrophil, eosinophil, basophil.

Leukocytosis

A white cell count above normal (above about 11,000 per microliter), usually from infection or inflammation.

Diapedesis

A white cell squeezing out of a capillary between the endothelial cells to reach the tissue.

-philia and -cytosis

Word endings that mean too many of that cell, as in neutrophilia or lymphocytosis.

Anemia

Too little oxygen-carrying capacity: too few red cells or too little hemoglobin.

Polycythemia

Too many red cells, which thickens the blood.

Hemophilia

An inherited lack of a clotting factor, so blood clots slowly.

-penia

A word ending that means too few of that cell, as in neutropenia or thrombocytopenia.

Sickle cell disease

An inherited change in hemoglobin that bends red cells into a sickle shape when oxygen is low.

Leukemia

A cancer of white cell-forming tissue that fills the marrow with abnormal white cells.

Rouleaux

Red cells stacked like a roll of coins, seen on some smears and in small vessels.

Questions you could be asked on the lab exam.

At each station a structure will be pinned or pointed to. These are the kinds of questions that go with it. Practice answering them out loud with a partner, with the structure in front of you.

Muscle tissue and microanatomy

1. Identify the tissue on this slide: skeletal, cardiac, or smooth muscle.
2. Name the structure at the pointer (for example, intercalated disc, peripheral nucleus, or striation).
3. Is this skeletal muscle cut longitudinally or in cross section?
4. Name the connective tissue layer at the pointer: epimysium, perimysium, or endomysium.
5. What does the connective tissue layer at the pointer surround?
6. Name the band, line, or zone of the sarcomere at the pointer.
7. Which filaments are found in the region at the pointer: thick, thin, or both?
8. Name the structure on the muscle fiber model: T tubule, terminal cisterna, sarcoplasmic reticulum, or triad.
9. Which sarcomere regions get narrower when the muscle contracts, and which one stays the same?
10. Name the fascicle arrangement of this muscle and give another muscle with the same arrangement.
11. Name the class of lever shown and label the fulcrum, effort, and load.

Muscles of the head and neck

1. Identify the muscle at the pointer.
2. What expression does the muscle at the pointer produce?
3. Which cranial nerve supplies the muscle at the pointer?
4. Is the muscle at the pointer a muscle of facial expression or a muscle of mastication?
5. Where does the muscle at the pointer insert?
6. Is the muscle at the pointer suprahyoid or infrahyoid, and does it raise or lower the hyoid?
7. Name the muscle that divides the neck into its anterior and posterior triangles.
8. What happens to the head when only the left sternocleidomastoid contracts?

Muscles of the back and thorax

1. Identify the muscle at the pointer.
2. Give the origin, insertion, action, or nerve of the muscle at the pointer.
3. Name the three erector spinae columns from lateral to medial, and identify the one at the pointer.
4. Which direction do the fibers of the external intercostals run, and what do they do to the ribs?
5. What structures pass through the caval opening, the esophageal hiatus, and the aortic hiatus, and at which vertebral levels?
6. Which nerve supplies the diaphragm, and from which spinal segments?
7. Which nerve supplies serratus anterior, and what sign follows its injury?
8. Name the muscle that attaches to the coracoid process at the pointer.

Muscles of the upper extremity

1. Identify the muscle at the pointer.
2. Give the origin, insertion, action, or nerve of the muscle at the pointer.
3. Which rotator cuff muscle attaches to the lesser tubercle?
4. Name the tubercle this rotator cuff muscle reaches and the movement it produces.
5. Is this forearm muscle in the anterior or posterior compartment, and which nerve supplies this muscle?
6. Name the common origin shared by the muscles at the pointer: medial or lateral epicondyle.
7. Which muscle flexes the elbow in every forearm position, and which is the prime mover of pronation?
8. Name the landmark at the pointer: cubital fossa, carpal tunnel, flexor retinaculum, or anatomical snuffbox.
9. Is the muscle at the pointer thenar, hypothenar, or midpalmar?
10. What do the palmar interossei do, and what do the dorsal interossei do?

The heart

1. Identify the chamber, valve, or vessel at the pointer.
2. What two chambers, or which chamber and vessel, does this valve separate?
3. Name the layer of the heart wall or pericardium at the pointer.
4. Identify the internal feature at the pointer: pectinate muscles, trabeculae carneae, papillary muscle, chordae tendineae, fossa ovalis, or moderator band.
5. Name the coronary artery or cardiac vein at the pointer and the sulcus it runs in.
6. Where does the coronary sinus empty?
7. Trace a drop of blood from the superior vena cava to the aorta, naming every chamber and valve.
8. Name the part of the conduction system at the pointer.
9. What fetal structure became the ligamentum arteriosum, and what did it connect?

Blood vessels

1. Is this vessel an artery or a vein? Name one feature you used to decide.
2. Name the tunic or layer at the pointer.
3. Is this an elastic artery, a muscular artery, or an arteriole?
4. Name the capillary type shown and one organ where it is found.
5. Identify the named artery or vein at the pointer.
6. At which landmark does the subclavian artery become the axillary artery? The axillary become the brachial?
7. Name the three branches of the aortic arch in order from right to left.
8. Name the vein most often used to draw blood and where it lies.
9. Name the adult remnant of the fetal structure at the pointer.

Nerves of the upper limb

1. Identify the part of the brachial plexus at the pointer: root, trunk, division, cord, or branch.
2. Name the terminal nerve at the pointer and the cord it comes from.
3. Which compartment or muscles does this nerve supply?
4. Name the nerve injured at this site and the sign it produces (for example, wrist drop or claw hand).
5. Which nerve runs behind the medial epicondyle?
6. Which nerve passes through the carpal tunnel?

Blood

1. Identify the cell at the pointer.
2. Is the cell at the pointer a granulocyte or an agranulocyte?
3. Is the cell at the pointer from the myeloid line or the lymphoid line?
4. What is the main job of the cell at the pointer?
5. Name the layer at the pointer in the spun tube: plasma, buffy coat, or red cells.
6. What does the hematocrit measure?
7. Put the five white cells in order from most to least numerous.
8. Which white cell would you expect to be high in a parasitic worm infection? In an acute bacterial infection? In mononucleosis?
9. What is the term for too few platelets? Too many white cells?
10. Name the cell that breaks apart to form platelets.